Probability 1 (MATH 11300) sample class – problem sheet

February 2021

Answers

Questions marked * are to be handed in.

*1. You arrive at a bus stop at 10 o'clock, knowing that the bus will arrive at some time uniformly distributed between 10 and 10:30.

(a) What is the probability that you will have to wait longer than 10 minutes?

(b) If at 10:15 the bus has not yet arrived, what is the probability that you will have to wait at least an additional 10 minutes?

Answer:

(a) 2/3.

(b) 1/3. Remark: in the case of uniform waiting times, conditioned on the event we wait for not having happened yet, the remaining waiting time is again uniform (on the remaining interval as the set of possible values).

2 A bus travels between the two cities A and B, which are 100 km apart. If the bus has a breakdown, the distance from the breakdown to city A has a uniform distribution over (0, 100). There is a bus service station in city A, in B, and at half way on the route between A and B. It is suggested that it would be more efficient to have the three stations located 25, 50, and 75 km, respectively, from A. Do you agree? Why? What would be the optimal location of the three stations?

Answer: First of all, we have to clear what the word "efficient" means in this setting. A good interpretation can be that the expectation of the distance D from the location of the breakdown to the nearest service station should be minimal. Let the three service stations be in distances $a < b < c$ from A. Then

$$D = \begin{cases} a - X & \text{, if } 0 < X < a, \\ X - a & \text{, if } a < X < [a+b]/2, \\ b - X & \text{, if } [a+b]/2 < X < b, \\ X - b & \text{, if } b < X < [b+c]/2, \\ c - X & \text{, if } [b+c]/2 < X < c, \\ X - c & \text{, if } c < X < 100. \end{cases}$$

X is uniform, its density is $1/100$ between the two cities, and so

$$\begin{aligned} \mathbb{E}(D) = \frac{1}{100} \cdot \Big[& \int_0^a (a-x)\,\mathrm{d}x + \int_a^{\frac{a+b}{2}} (x-a)\,\mathrm{d}x + \int_{\frac{a+b}{2}}^b (b-x)\,\mathrm{d}x \\ & + \int_b^{\frac{b+c}{2}} (x-b)\,\mathrm{d}x + \int_{\frac{b+c}{2}}^c (c-x)\,\mathrm{d}x + \int_c^{100} (x-c)\,\mathrm{d}x \Big] \\ = \frac{1}{100} \cdot \Big[& \frac{a^2}{2} + \frac{(b-a)^2}{4} + \frac{(c-b)^2}{4} + \frac{(100-c)^2}{2} \Big]. \end{aligned}$$

In the current situation $a = 0,\ b = 50,\ c = 100$, and $\mathbb{E}(D) = 12.5$ kilometers. In the suggested arrangement $a = 25,\ b = 50,\ c = 75$, and $\mathbb{E}(D) = 9.375$ kilometers, which means it is more efficient than the current configuration. It is not difficult to see that the optimal arrangement (the one that minimizes $\mathbb{E}(D)$) is $a = 50/3 \simeq 16.7,\ b = 50,\ c = 250/3 \simeq 83.3$.

It is a little bit more involved to argue that the suggested arrangement $a = 25,\ b = 50,\ c = 75$ is *stochastically better* than the current configuration that is, for any d, $\mathbb{P}\{D < d\}$ is at least as large in the suggested setting than in the current setting. It is also possible to see that the arrangement $a = 50/3,\ b = 50,\ c = 250/3$ that optimizes $\mathbb{E}(D)$ is *stochastically optimal* that is, it maximizes $\mathbb{P}\{D < d\}$ for each d at the same time. Therefore the suggested arrangement is not only better in terms of $\mathbb{E}(D)$ than the current configuration, and the third arrangement not only optimizes $\mathbb{E}(D)$, but the suggested arrangement gives a smaller $\mathbb{E}(g(D))$ than the present configuration, and the third arrangement optimizes $\mathbb{E}(g(D))$ for any non-decreasing function $g(D)$ of D. For example, the suggested setting gives a smaller expected distance square $\mathbb{E}(D^2)$, and the third arrangement optimizes $\mathbb{E}(D^2)$ as well.

These statements follow because the problem is relatively simple. In more complicated situations there does not have to be an optimal solution. It is possible that some arrangement will minimize $\mathbb{E}(D)$ for some quantity D of interest, but another arrangement might minimize for example $\mathbb{E}(D^2)$. In this case one needs to clear what is meant by "good" and "better" configurations.